

LECTURE NO 10: *Sustainability, Ethics, and Technology*

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1.1 Introduction (1/5)

Case Biofuels



Figure 1: Biofuel. Photo: © Hajohoos / Fotolia.com.

1.1 Introduction (2/5)

How can we continue to provide the energy for our transport needs? According to a theory known as “Peak Oil,” the production of oil will decline from now on. At the meantime, our transportation needs seem to continue growing. This scenario implies that we cannot continue to rely on oil for transportation. An additional reason for aiming at a reduced or no reliance on oil for transportation is the substantial contribution that the use of oil makes to the greenhouse effect through car emissions.

One proposed approach to this problem is to develop and use biofuels. Biofuel is solid, liquid, or gaseous fuel obtained from relatively recently lifeless or living biological material. The main difference with fossil fuels is that the latter are derived from long dead biological material. The use of biofuel appears to have two obvious advantages. First, we will not run out of fuels since we can always grow more. Second, the plants that are used to make biofuels extract the greenhouse gas CO_2 from the atmosphere, so reducing the greenhouse effect. The proverbial two birds with one stone.

1.1 Introduction (3/5)

To start with its contribution to global warming. The reasoning above ignores the fact that farming costs energy because of the machinery used, and because of the fertilizers that need to be produced. Turning crops into biofuels also consumes energy. Given that this is energy spent “outside” the cycle of letting the plants grow and using the end products as fuel, it is obvious that the contribution to global warming is not as minimal as it would first seem. Perhaps more important are the unwanted side effects of using crops for producing fuel. Higher demand generally means higher prices. Crops such as sugarcane, wheat, and corn are likely to become more expensive which could well mean that those that are already struggling to feed themselves will no longer manage.¹

1.1 Introduction (4/5)

Sometimes it is claimed that second generation biofuels solve these problems. Second generation biofuels are based on non-food crops. However, producing such crops still requires land, water, and fertilizer, which may become short in supply or the prices of which will drastically increase. Therefore, second generation biofuels are likely to contribute to an indirect increase in food prices. Moreover, the drive to produce large amounts of such crops is likely to reinforce already existing negative trends such as deforestation etc.

Third generation biofuels – producing fuels by means of bacteria or algae – are being researched and developed in answer to these concerns. However, while they fare substantially better on the above points than first- or second-generation biofuels, they are currently prohibitively expensive to produce.

Source: Based on Naylor et al. (2007); Inderwildi and King (2009); Zah et al. (2007).

1.1 Introduction (5/5)

- A number of questions emerge from the case study- *Biofuels* . Do we have an obligation to prevent future environmental problems, even if they mostly affect future generations? Isn't it better to leave every generation to solve the problems that are relevant to their lifetime?.
- Moreover, the discussion about biofuels clearly shows that there is a trade-off between our obligations to future generations and the present one.
- Climate change, deforestation, air pollution, soil contamination, the availability of clean water, and overfishing are just a few examples of environmental problems that have increasingly drawn the attention and concern of both governments and individuals over the past few years. In many cases, technology is part of the cause of these problems but often it is also potentially a part of the solution. In philosophical ethics, these problems have led to attention for what is called “*environmental ethics*.”

1.2 Environmental Ethics? (1/2)

- Ethics is first and foremost involved with interaction between people. Even though you may have come across the notion that people have a certain responsibility regarding the environment, is this in fact morally justifiable?
- In the first instance two different answers can be given to this. *First answer*, you could say that responsibility for the environment is derived from responsibility for the humankind and society. For if we pollute the environment at will and exhaust our resources burn up all our fossil fuels now – there may be severe consequences for people now and in the future. In environmental ethics this kind of justification is known as *anthropocentrism* ; it is a position that states that the environment only has an *instrumental value*, that is, the value of its use by us.
- A healthy environment is necessary to provide the essential basic needs, such as food, clean air, and clean water.

1.2 Environmental Ethics? (2/2)

- The *second answer* argues that the environment has a value of its own (an *intrinsic value*) and, therefore, should be considered in moral arguments. This is known as *biocentrism*.
- Even if the environment (or part of it) does not have any use value (*instrumental value*), it still has moral value.
- If the environment has moral value, be it instrumental or intrinsic, it deserves to be protected. From this it follows that people in general are responsible for the environment. Moreover, engineers have a special responsibility, because technology has both positive and negative effects on the environment.
- There is much consensus in society concerning our environmental problems and that environmental care is of importance, and moreover that technology can play an important role in this.

1.3 Environmental Problems? (1/2)

- There are many kinds of environmental problems, and solving them first requires their precise definition. In general three kinds of environmental problems are distinguished:
 1. **Pollution**- Environmental problems in which something undesirable or damaging is added to the environment.
 2. **Exhaustion**- A type of environmental problem in which something valuable is removed from the environment that cannot, or at least not easily, be renewed. Fossil fuel is a non-renewable source, biofuel is a renewable source.
 1. **Non-renewable resources**- Natural resources that cannot be renewed or reproduced. An example is fossil fuel.
 2. **Renewable resources**- Natural resources that can be renewed or reproduced. An example is biofuel.
 3. **Degradation**- Structural damage to the environment. An example is soil erosion.

1.3 Environmental Problems? (2/2)

- We do know that the continuity of human welfare and well-being is endangered by exhausting resources, and that polluting the water, air, and soil means we have to make increasing efforts to keep the environment liveable. As humans we depend on the environment.
- Environmental problems also have an important social dimension. For example, housing near to sources of pollution is cheaper, such as housing near to highways, municipal dumps, major industrial installations, etc.

1.4 Sustainable Development

1.4.1 *The Brundtland definition*

- The best-known definition of sustainable development originates from the Brundtland report.
- *Sustainable development*- Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs (Brundtland definition).

1.4.2 *Moral justification*

- The heart of sustainable development lies in two kinds of justice.
 1. *Intergenerational justice*: Justice that relates to the just distribution of resources between different generations.
 2. *Intragenerational justice*: Justice that relates to the just distribution of resources within a generation.

1.5 Environmental Space

- *Environmental space*- The (maximum) amount of use of renewable and non-renewable resources that does not exceed the boundaries of what the environment can take.
- *Carrying capacity*- The amount of damage that can be done to the environment without that damage being irreversible.

1.6 Can a Sustainable Society be Realized? (1 / 3)

- It is often said that a sustainable society simply cannot be realized. “Sustainable development is not yet achievable, because it is not cost effective economically speaking,” or “consumers just don’t want to pay more for sustainable products.”
- If we say that something is possible or impossible, we always have certain boundary conditions in mind. Without boundary conditions it is meaningless to speak about possibilities. And these boundary conditions can be debated too; “I couldn’t finish the assignment because I had promised my friends to meet them at the cinema” is a less acceptable excuse compared to, “I couldn’t finish the assignment because I had to take my mother to hospital.”

1.6 Can a Sustainable Society be Realized? (2/3)

- When we say why a sustainable society is not achievable, we often are confronted with economic boundary conditions. “It is impossible to fulfil our transportation needs by means of third-generation biofuels, since they are too expensive.”
- Some sustainable developments are socially unachievable. Not many people would be willing to reduce their present mobility to reduce harm to the environment—which would be the morally desirable thing to do. We want to keep going on holiday to other parts of the world and we want to go to work with our own car. Our mobility has become so important that we do not want to sacrifice it as a contribution to a more sustainable society.
- Sometimes there are boundary conditions with strong arguments in their favour that make the sustainable option impossible. But it is a good thing to make this explicit, rather than terminating all discussions by saying “this is impossible!” without supplying reasons.

1.6 Can a Sustainable Society be Realized? (3/3)

- Politicians are unlikely to take the drastic measures that are needed for sustainability.
- There are no experts — either of technology or environmental care — that can give us the complete story about how to achieve sustainability. Even if their factual knowledge is great, their expertise and authority is no greater than that of an ordinary person.
- Would a green dictatorship be an option then? Even if a dictator had the best intentions regarding the realization of sustainability, we have often seen that dictatorships fail to achieve many of their original aims. Thus there is little to be expected from that solution.
- Some believe that sustainable technologies will arise because of the demand. So the market will bring a solution, as it has in many other instances.
- By waiting for the market to solve the problem, it is possible that a worse situation will arise for everyone.

1.7 Engineers and Sustainability (1 / 2)

- Although there is only limited agreement on what a sustainable society is, and on how or whether it will be achieved, engineers can certainly make a contribution. Technical knowledge can be used in different ways to directly or indirectly solve or prevent environmental problems. This means engineers are responsible for making use of those opportunities.
- Various technical measures can be taken to satisfy this responsibility for the environment. These measures can be categorized according to level and type. The levels are as follows:
 - a. Product level;
 - b. Process level;
 - c. Business level.

} Measures according to levels

The types are as follows:

1. Cleaning up pollution;
2. Processing of waste flows;
3. Preventing waste flows.

} Measures according to types

1.7 Engineers and Sustainability (2/2)

1.7.1 Points of attention during the design process

- As we saw in Lecture 7 (*Ethical questions in the design of technology*), a number of moral questions can be asked during the design process and this also applies to sustainability. These questions can concern the environmental impact of a product during its *life phases*.
- **Life phases-** The phases through which a product goes during its “life”: *production phase*, *use phase*, and *removal phase*. Each phase raises its own environmental questions. The choices for the environmental impact of one phase can influence the environmental impact of another phase.

1.7.2 Life cycle analysis- An analysis that maps the environmental impact of a product across the entire cycle of production, use, and disposal.

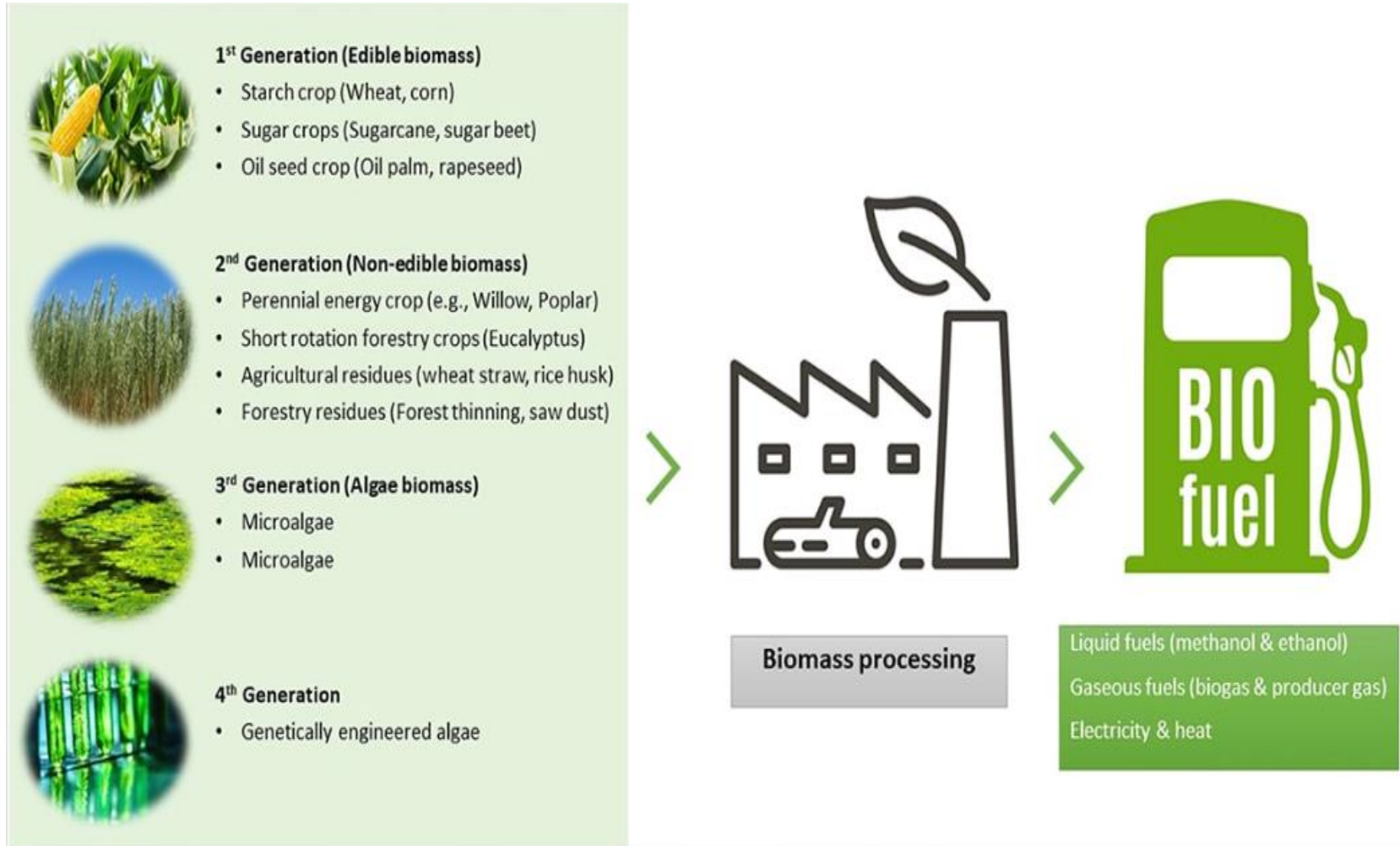
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Supplementary Material

1.8 Supplementary Material (1/3)

Biofuels Generations



1.8 Supplementary Material (2/3)

- **Biofuel:** Any fuel that is derived from biomass- that is plant or algae material or animal waste. Biofuel is considered to be a source of renewable energy, unlike fossil fuels such as petroleum, coal and natural gas.
- **Algae:** are organisms or living things, that are found all over the world. However algae are actually neither plants nor animals. Instead they belong to a group of living things called *protists*.
- **Soil Erosion:** It is the displacement of the upper layer of soil; it is a form of soil degradation. This natural process is caused by dynamic activity of erosive agents, that is water, ice, snow, air, plants and humans.

1.8 Supplementary Material (3/3)

Intrinsic vs. Instrumental Values

- ❑ Philosophers distinguish between two types of values: *intrinsic* and *instrumental* values.
- ❑ Any value that serves some further end or good is called an instrumental value because it is tied to some external standard.
- ❑ Automobiles, computers, and money are goods that have instrumental value.
- ❑ Values such as life and happiness are intrinsic because they are valued for their own sake.